

Question Number	Answer	Additional Guidance	Mark
2(a)(i)	An answer that includes the following three points. - the water content is reduced (1) - the seeds are x-rayed (1) - the seeds are kept at {low temperature / dry} (1)	Accept seeds dried Accept low oxygen environment	(3)

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2(a)(ii)	An answer that includes the following points. - the {number of different / variety of} species (of living organisms) in an area / stated area (1) - the genetic differences within species of living organisms (1)	Accept species richness of an area Accept genetic variety	(2)

Question Number	Answer	Additional Guidance	Mark
2(b)(i)	A description that includes 4 of the following points: - fixing root tip described (1) - stain used (1) - description of maceration / squashing (1) - view through high power microscope (1) - count of number of dividing cells and divide by total number of cells (1)	e.g. heating with conc. HCl Accept magnification quoted $\geq 100\times$	(4)

Question Number	Answer	Additional Guidance	Mark
2(b)(ii)	An answer that includes the following points. • read MI from graph (1) • rearrange equation $MI = \text{no. of cells in mitosis} \div \text{total number of cells}$ to calculate total number of cells (1) • substitute values in this equation and correct answer (1)	e.g. 5(%) total number of cells = no. of cells in mitosis $\div$ MI total no. of cells = $14 \div 0.05 = 280$ / T = $1400 \div 5 = 280$ correct answer with no working gains 3 accept T = 14 / 5 for 2	(3)

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2(b)(iii)	A table drawn showing: • suitable table (drawn) (1) • headings root length with units, mitotic index with units (1) • data correctly entered (1)	2 columns and 6 rows or vice versa units in cells negates this mark <table><thead><tr><th>Root Length / mm</th><th>Mitotic Index (%)</th></tr></thead><tbody><tr><td>5(.0)</td><td>0.0</td></tr><tr><td>6(.0)</td><td>0.4</td></tr><tr><td>8(.0)</td><td>0.8</td></tr><tr><td>10(.0)</td><td>1.4</td></tr><tr><td>12(.0)</td><td>1.8</td></tr></tbody></table>	Root Length / mm	Mitotic Index (%)	5(.0)	0.0	6(.0)	0.4	8(.0)	0.8	10(.0)	1.4	12(.0)	1.8	(3)
Root Length / mm	Mitotic Index (%)														
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2(b)(iv)	A description which includes the following points: - the {treated / aged} seeds have a lower mitotic index than the untreated seeds (1) - the longer the treatment the more the mitotic index is reduced / 14 days more than 4 days (1) {optimum / maximum} mitosis occurs in longer roots with greater aging (1)	Accept mitosis starts later in more aged roots	(3)

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2(b)(v)	A description which includes the following points: - experiment repeated (1) - under the same conditions / described (1)		(2)